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DETAILED RESEARCH PROPOSAL

I. GENERAL INFORMATION

Research Title:

MedCore: An AI-powered platform for transparent Hospital Billing and Insurance Reconciliation

Proponent(s):

CALLUENG, MARC ANDRE NICOLAS M.

MONTENEGRO, MICOH JEREMY C.

FERNANDO M. JAMINOLA JR.

ONZA, MCLEIN JEI G.

Project Study Leader:

FERNANDO M. JAMINOLA JR.

Co-Researcher(s):

CALLUENG, MARC ANDRE NICOLAS M.

MONTENEGRO, MICOH JEREMY C.

ONZA, MCLEIN JEI G.

Implementing Institution(s):

Cagayan State University – Carig Campus, College of Information and Computing Sciences

Cooperating Agency(ies):

Regional/Local partner hospitals and healthcare facilities within Cagayan Valley (Region II) for pilot testing and validation.

Duration:

12 Months (Start Date: November 3, 2026 – Completion Date: November 3, 2027)



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II. INTRODUCTION

A. Background/Rationale:

Hospital financial administration in the Philippines is severely constrained by legacy database architectures and disconnected, manual administrative calculations. Under the Philippine healthcare financing landscape, hospital care operates within a fragmented multi-payer environment comprising three divergent financial layers: on-premise Hospital Information Systems (HIS) managing unadjudicated chargemaster items and handwritten physician professional fee (PF) slips; the Philippine Health Insurance Corporation (PhilHealth), which provides foundational social health insurance under All Case Rates (ACR); and private Health Maintenance Organizations (HMOs) and commercial insurers regulated by the Insurance Commission under Executive Order No. 192 (s. 2015). While the Universal Health Care (UHC) Act (RA 11223) envisions comprehensive financial protection, official Philippine Statistics Authority (PSA, 2024) data indicates that Current Health Expenditure reached ₱1.56 Trillion, with catastrophic Household Out-of-Pocket (OOP) expenses accounting for 42.7% (₱615 Billion). Concurrently, private Voluntary Health Insurance (VHI) and commercial HMOs have expanded rapidly to represent 13.0% (₱161.35 Billion) of national health expenditures, disbursing over ₱61.05 Billion in healthcare claims and benefits in 2024 across more than seven million enrolled corporate members. In the absence of an integrated reconciliation middleware, hospital billing departments, patients, and private insurers operate in informational silos, resulting in pervasive billing opacity and prolonged discharge bottlenecks.

When an inpatient is certified as clinically cleared for discharge through an attending physician's 'May Go Home' (MGH) order, the administrative clearance process routinely stalls. In Philippine hospitals, discharge turnaround time (TAT) ranges between 2.5 to 4.5 hours in community facilities and escalates to 4.0 to 8.0+ hours in tertiary private medical centers. Empirical operations research demonstrates that the single largest delay factor occurs during private HMO Letter of Authorization (LOA) adjudication, consuming 2.0 to 4.0 hours per patient. Hospital billing clerks must manually transcribe paper charges into disparate insurer web portals or clearinghouses, awaiting off-site claims adjusters who routinely disallow routine hospital supplies (e.g., PPE, admission kits, cannulas, and ward sundries) and enforce strict room-and-board inner limits. Compounding this friction is the contractual 'PhilHealth-First' rule: HMO policies strictly require the statutory PhilHealth case rate to be calculated and deducted before private coverage applies. Any clerical delay or missing documentation in PhilHealth verification immediately halts HMO processing, shifting the unfiled deduction straight into the patient's out-of-pocket cash obligation. In Philippine society, women disproportionately carry the burden of family caregiving and front-desk financial coordination, enduring hours of anxiety and opaque cashier confrontations. Furthermore, over 75% of frontline hospital billing staff, nursing coordinators, and social workers are women who absorb the emotional distress and unpaid overtime caused by these systemic administrative failures.



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MedCore directly resolves this multi-payer crisis by introducing a non-invasive, intelligent financial reconciliation and multi-payer middleware platform sitting between legacy hospital systems, PhilHealth, private HMOs, and patients. Operating through a closed-loop 5-pillar value engine (GUIDE, VERIFY, REWARD/SETTLE, REPORT, and IMPROVE), MedCore non-invasively captures real-time hospital chargemaster feeds, automates the complete statutory deduction cascade (12% VAT exemption, 20% Senior Citizen and PWD discounts under RA 9994/RA 10754, PhilHealth ACR splits, and private HMO eLOA benefit caps), and compresses discharge workflows toward an operational SLA target of under 45 minutes. By decoupling deterministic billing mathematics from privacy-preserving AI proxies (Google AI Studio Gemini 2.5 Flash with automated Protected Health Information de-identification under RA 10173), MedCore provides patients with real-time, plain-language billing explanations via SMS magic links while enforcing statutory protections such as legally binding 0% interest digital promissory notes under Republic Act No. 9439 (Anti-Hospital Detention Law).

B. Statement of the Problem/Objectives:

General Objective:

Generally, this study aims to design, develop, and evaluate MedCore, an intelligent multi-payer middleware platform engineered to streamline hospital billing transparency, automate statutory and private HMO reconciliation, and accelerate digital patient discharge workflows across Philippine healthcare facilities. Specifically, this research seeks to answer the following questions:

1. **1.** What is the current baseline hospital discharge turnaround time (TAT) and bed turnover time across partner healthcare facilities, and what are the specific delay contributions attributable to attending physician professional fee posting, PhilHealth eClaims compilation, and private HMO final Letter of Authorization (eLOA) adjudication?
2. **2.** How can the MedCore middleware platform be architecturally designed and developed to: (a) non-invasively ingest unadjudicated chargemaster and encounter feeds from legacy hospital information systems (HIS); (b) automate the multi-tier statutory deduction cascade, coordinating 12% VAT exemptions, 20% Senior Citizen and PWD discounts (RA 9994/RA 10754), PhilHealth All Case Rates (ACR), and private HMO benefit schedules; (c) operationalize the 5-pillar closed-loop workflow (GUIDE, VERIFY, RECONCILE, REWARD/SETTLE, REPORT, IMPROVE) to compress discharge turnaround time below 45 minutes; (d) deploy privacy-preserving Artificial Intelligence modules—including 'Cora', the multilingual patient billing concierge chatbot, clinical claims forensic pre-flight scrubbing for PhilHealth RTH risk prediction, automated HMO medical necessity appeal drafting, and executive bottleneck telemetry; and (e) streamline commercial HMO electronic Letter of Authorization (eLOA) generation and PhilHealth eClaims 3.0 eSOA XML compliance?



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3. The Technology Acceptance Model (TAM) will be used to evaluate platform adoption across two primary constructs: Perceived Usefulness (how the automated statutory calculation engine accelerates workflows) and Perceived Ease of Use (how easily patients navigate SMS magic links and digital interfaces). Concurrently, the ISSM provides the theoretical basis for evaluating the system's technical rigor. System Quality will be measured by the reliability of the database connector to legacy HIS/EMRs. Information Quality will evaluate the centavo-exact accuracy of the multi-tier deductions (12% VAT, SC/PWD, PhilHealth ACR, and HMO policy limits). Finally, Net Benefits serves as the framework for measuring the overall reduction in hospital discharge turnaround times and reclaimed bed capacity.

4. **4.** Is there a statistically significant reduction in hospital discharge turnaround time (TAT) and hospital bed turnover time after the implementation of the MedCore platform compared to the established pre-implementation baseline?

Objectives of the Study:

To answer the aforementioned questions, the researchers will pursue the following specific objectives:

1. **1.** To conduct a baseline time-motion study and needs assessment across partner hospital facilities to map current discharge bottlenecks, quantifying the specific time sinks incurred by manual doctor PF posting, PhilHealth claims verification, and private HMO final LOA adjudication.
2. **2.** To design and develop the MedCore multi-payer middleware, encompassing the non-invasive legacy HIS connector, the automated multi-tier statutory deduction engine, the 5-pillar closed-loop operating architecture (GUIDE pre-admission orientation, VERIFY real-time coverage, REWARD/SETTLE rapid clearance and digital promissory notes, REPORT telemetry and eClaims XML auditing, and IMPROVE denial learning), the commercial HMO eLOA coordination module, and the privacy-preserving AI suite featuring the 'Cora' multilingual patient billing concierge, clinical claims forensic scrubber, automated HMO appeal generator, and executive command briefings.
3. **3.** To evaluate the technical software quality using the ISO/IEC 25010 standard with IT experts and administrators, assess user acceptability using the Technology Acceptance Model (TAM) with hospital personnel and patients, and measure gender-responsiveness using the Harmonized Gender and Development Guidelines (HGDG).
4. **4.** To measure and evaluate the platform's real-world operational impact by statistically comparing post-implementation hospital discharge turnaround times and bed turnover rates against established baseline metrics.



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III. REVIEW OF LITERATURE:

Healthcare financial administration in the Philippines operates within a highly fragmented multi-payer environment characterized by severe out-of-pocket exposure and divergent institutional payment pipelines. Official data from the Philippine Statistics Authority (PSA, 2024) Philippine National Health Accounts reveals that total Current Health Expenditure (CHE) expanded to ₱1.56 Trillion, with Household Out-of-Pocket (OOP) financing representing a dominant 42.7% (₱615 Billion). While the Universal Health Care (UHC) Act (Republic Act No. 11223) mandates automatic entitlement to the National Health Insurance Program, recent health economics evaluations demonstrate that social health insurance provides an effective financial support value of only 20% to 40% for inpatient confinements, leaving patients exposed to substantial unexpected expenses (Pantig & Ulep, 2025; Casas et al., 2025). To bridge this coverage shortfall, private Voluntary Health Insurance (VHI) and commercial Health Maintenance Organizations (HMOs) have surged by 44.6% year-on-year to represent 13.0% (₱161.35 Billion) of national health expenditures, disbursing ₱61.05 Billion in medical benefits in 2024 (Insurance Commission, 2024). Pantig and Ulep (2025) emphasize that the structural lack of coordination between PhilHealth, commercial HMOs, and hospital providers generates severe administrative friction, opaque cost-shifting, and redundant billing procedures that undermine national universal healthcare objectives.

Health Maintenance Organizations in the Philippines are supervised and regulated by the Insurance Commission pursuant to Executive Order No. 192 (Series of 2015). Under rigorous capitalization and solvency requirements, the domestic HMO sector has consolidated into 25 licensed entities—led by major corporate providers including Maxicare Healthcare Corporation, Intellicare (Asalus Corporation), Medicard Philippines, PhilCare, and Pacific Cross—covering over seven million corporate and individual subscribers (Insurance Commission, 2024). To manage underwriting risk and loss ratios, commercial HMOs enforce intricate contractual provisions, including Maximum Benefit Limits (MBL), strict room-and-board daily allowances, and exhaustive consumable exclusion lists that disallow standard hospital sundries (such as PPE, admission kits, cannulas, and routine ward supplies) as non-payable overhead (Insurance Commission, 2024; MedCore Vault Domain Analysis). In hospital operations, the inpatient discharge clearance process represents the single largest non-clinical delay, requiring 150 to 270 minutes (2.5 to 4.5 hours) in community facilities and 240 to 480+ minutes (4.0 to 8.0+ hours) in tertiary private institutions (Desalva et al., 2021). Operational analyses establish that manual HMO Letter of Authorization (LOA) adjudication is the primary culprit, consuming 2.0 to 4.0 hours per patient as hospital clerks manually assemble preliminary statements of account, transmit them to remote corporate claims examiners, and await line-by-line manual scrubbing. Struck-out expenses are abruptly shifted into the patient's personal balance at the cashier desk, driving a 33.8% surge in formal consumer disputes handled by the Insurance Commission's Public Assistance and Mediation Division (PAMD, 2023).



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A critical systemic bottleneck identified in Philippine health financing is the universal 'PhilHealth-First' coordination rule enforced across commercial HMO contracts. Under this policy, private HMO benefits are strictly secondary and supplemental to state health insurance; hospitals must calculate and deduct the statutory PhilHealth case rate before HMO coverage can be applied against the remaining bill. When a patient encounters administrative delays in securing their PhilHealth Member Data Record (MDR), qualifying contribution records, or Claim Form 1 (CF1), HMO claims examiners automatically deduct the estimated PhilHealth rate from their approved authorization, converting it into an immediate, out-of-pocket cash liability for the patient. As documented by Banogon et al. (2025) and Co et al. (2026), hospital billing staff are forced to operate as human clearinghouses, toggling between disconnected software terminals: the hospital's on-premise Hospital Information System (e.g., BizBox, Comlogik, Medsys), the PhilHealth portal, and multiple standalone HMO portals or clearinghouse terminals. This fragmented re-keying creates clerical errors, billing disputes, and Return-to-Hospital (RTH) claim rejections of 15% to 20%, trapping between ₱4 Billion and ₱10 Billion in unpaid claims that severely constrain private hospital liquidity (Private Hospitals Association of the Philippines [PHAPi], 2024).

To modernize provider reimbursement, PhilHealth has rolled out standardized electronic claims protocols, including PhilHealth Circular No. 2019-0002 mandating electronic Claim Form 4 (CF4) clinical abstracts and Circular No. 2023-0026 mandating machine-readable XML Statements of Account (eSOA) under eClaims 3.0. Scanned image or PDF billing statements have been permanently phased out in favor of strict mathematical validation against the national standard library. Furthermore, PhilHealth officially suspended and subsequently abolished the 90-day Single Period of Confinement (SPC) policy under Circular No. 2024-0021 (effective October 1, 2024), authorizing claims for repeated inpatient admissions and requiring healthcare providers to upgrade their billing middleware to support dynamic claims refiling and multi-encounter reconciliation. Hagag et al. (2025) and Casas et al. (2025) demonstrate that while these digital mandates modernize state claims processing, their success depends entirely on whether healthcare facilities can deploy intelligent middleware capable of harmonizing disparate hospital chargemasters with state and private insurer data schemas.

Hospital financial reconciliation in the Philippines is strictly governed by national statutes safeguarding patient welfare. Republic Act No. 9439 (Anti-Hospital Detention Law) prohibits the physical detention of medically cleared patients on grounds of nonpayment and establishes that patients in basic accommodations are entitled to depart upon executing a valid promissory note. Republic Act No. 10932 (Anti-Hospital Deposit Law) penalizes demanding advance payments for emergency care, while Republic Act No. 9994 and Republic Act No. 10754 mandate a sequential 12% VAT exemption followed by a 20% statutory discount for Senior Citizens and Persons with Disabilities. In parallel, Republic Act No. 10173 (Data Privacy Act of 2012) mandates rigorous end-to-end encryption and role-



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based access control for Protected Health Information (PHI). Crucially, health financing systems intersect directly with Gender and Development (GAD) mandates codified in Republic Act No. 7192, Republic Act No. 9710 (Magna Carta of Women), and the Harmonized Gender and Development Guidelines (HGDG, Third Edition, 2019 / NEDA, PCW, ODA-GAD Network). Under HGDG Box 13 (Health) and Box 19 (ICT), projects must evaluate gender-differentiated impacts. In the Philippines, women disproportionately carry the burden of informal caregiving, patient advocacy, and financial coordination during family hospitalizations, while over 75% of frontline hospital administrative personnel (nurses, billing clerks, and social workers) are women who absorb the emotional strain of hostile billing confrontations. Ensuring transparent, automated multi-payer adjudication and non-predatory digital promissory notes protects both vulnerable female caregivers and frontline healthcare workers.

International health informatics benchmarks, such as the Cedar Healthcare Platform and OODA Health reference architectures, demonstrate that addressing multi-payer fragmentation requires an intelligent middleware layer sitting non-invasively between legacy EHRs, state payers, and private insurers (Contrary Research, 2023). Rather than replacing complex hospital core systems, modern middleware utilizes automated Change Data Capture (CDC), deterministic calculation engines for centavo-exact statutory deductions, and privacy-preserving Artificial Intelligence to generate plain-language explanations. By integrating deterministic multi-payer rule engines with privacy-first Large Language Models (LLMs) that de-identify protected patient data prior to natural language synthesis, modern healthcare systems can compress discharge turnaround times from hours to minutes, eliminate billing disputes, and guarantee statutory compliance across public and private hospitals.

IV. CONCEPTUAL/THEORETICAL FRAMEWORK:

This study is anchored on the integration of the **Technology Acceptance Model (TAM)** by Fred Davis and the **Information Systems Success Model (ISSM)** by DeLone and McLean.

The Technology Acceptance Model (TAM) will be used to evaluate platform adoption across two primary constructs: Perceived Usefulness (how the automated statutory calculation engine accelerates workflows) and Perceived Ease of Use (how easily patients navigate SMS magic links and digital interfaces). Concurrently, the ISSM provides the theoretical basis for evaluating the system's technical rigor. System Quality will be measured by the reliability of the database connector to legacy HIS/EMRs. Information Quality will evaluate the centavo-exact accuracy of the multi-tier deductions (12% VAT, SC/PWD, PhilHealth ACR, and HMO policy limits). Finally, Net Benefits serves as the framework for measuring the overall reduction in hospital discharge turnaround times and reclaimed bed capacity.



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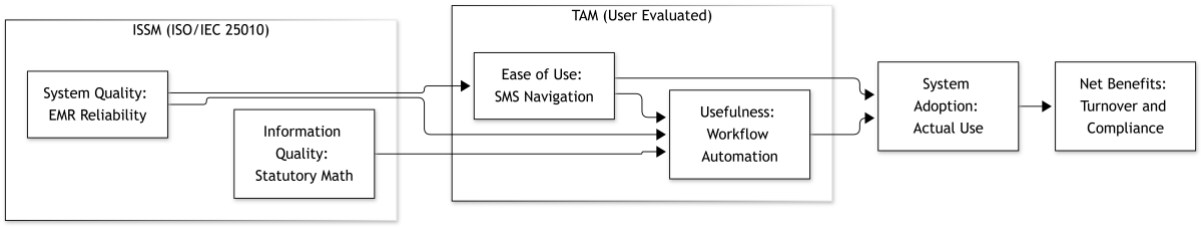


Figure 1. Theoretical Framework of the Study.

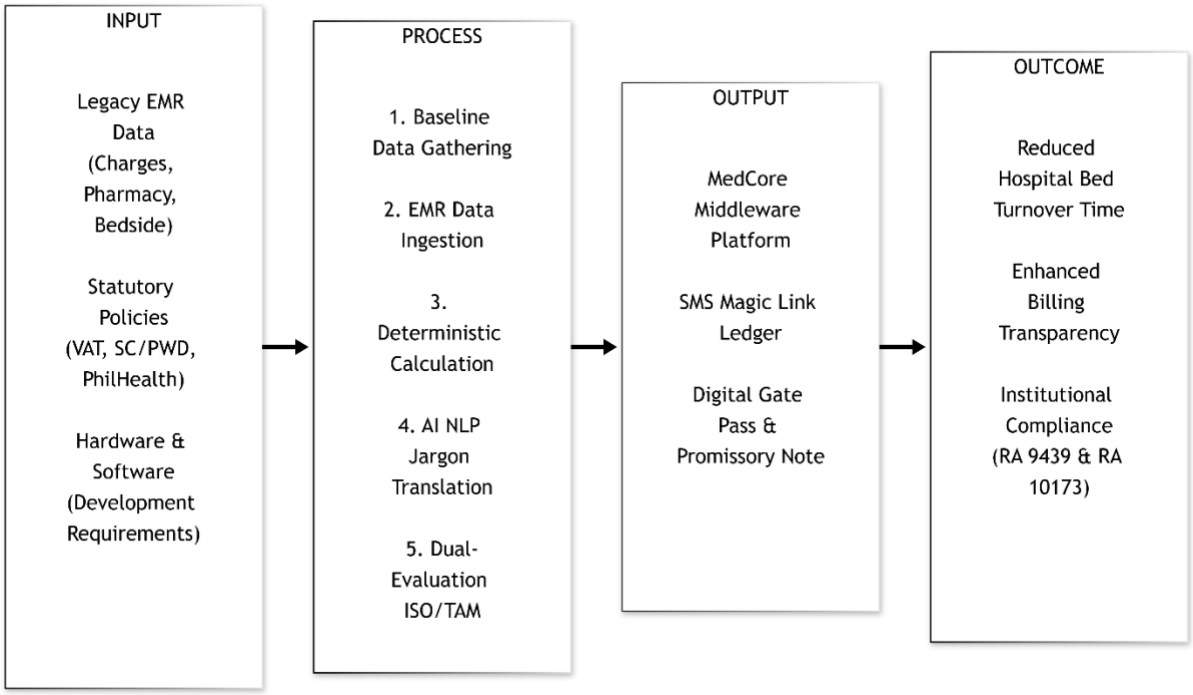


Figure 2. Conceptual Framework of the Study.

V. SIGNIFICANCE OF THE STUDY:

a) National Goals/Plans: Supports the Department of Health's (DOH) eHealth Strategic Framework, the Universal Health Care (UHC) Act (RA 11223), and the Insurance Commission's regulatory agenda under Executive Order No. 192 (s. 2015) and the Financial Products and Services Consumer Protection Act (RA 11765) by optimizing multi-payer healthcare financing, eliminating claims friction, and establishing transparent, coordinated provider payment mechanisms across state and commercial insurers.

b) National Policies: Ensures strict institutional compliance with Republic Act No. 9439 (Anti-Hospital Detention Law) through automated 0% interest digital promissory notes, Republic Act No. 10932 (Anti-Hospital Deposit Law) via non-binding pre-admission orientation, Republic Acts No. 9994 and 10754 through deterministic senior citizen and



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PWD discount calculations, Republic Act No. 10173 (Data Privacy Act of 2012) via role-based access and PHI anonymization, and Republic Act No. 9710 (Magna Carta of Women) and RA 7192 via Harmonized Gender and Development Guidelines (HGDG) gender-responsive standards.

c) Emerging Realities: Mitigates severe hospital administrative staffing shortages and multi-payer data fragmentation by replacing manual, phone-based HMO LOA chasing and multi-portal re-keying with automated tripartite reconciliation, PhilHealth eClaims 3.0 eSOA XML validation, and modern digital payment integrations (QR Ph, GCash, Maya).

d) Regional/Local Goals/Plans: Advances healthcare modernization across Cagayan Valley (Region II) by eliminating administrative discharge bottlenecks, directly improving hospital bed turnover rates, expanding clinical capacity, and aligning directly with the Provincial Government of Cagayan (PGC) EGAY Platform (Good Governance and Youth & Elderly Welfare pillars) while qualifying for statutory provincial 5% GAD budget attribution.

e) Community Goals/Plans: Protects financially vulnerable patients and female family caregivers from hidden out-of-pocket costs, arbitrary consumable disallowances, and traumatic discharge delays by delivering real-time, plain-language, transparent billing statements directly via mobile SMS magic links and providing humane, legally sound 0% interest repayment arrangements.

VI. PROCEDURE/METHODOLOGY:

The project will employ a participatory developmental approach involving the research project team and designated hospital personnel. The implementation will begin with consultation, needs assessment, and process mapping to identify requirements for the proposed system. These requirements will guide the design and architectural development of the MedCore platform, followed by comprehensive testing, user validation, iterative refinement, and pilot deployment within partner healthcare facilities.

After deployment, designated hospital personnel will undergo structured system orientation and hands-on operational training. The developed system will subsequently be evaluated by selected user cohorts in terms of technical software quality, operational usability, and clinical-administrative efficacy. User feedback and evaluation results will be utilized to identify necessary improvements and establish recommendations for the continued use and long-term sustainability of the system.

Research Design & Software Development Life Cycle (SDLC) The technical construction of MedCore will utilize the Evolutionary Prototyping SDLC model. Given the operational complexity of Philippine hospital billing workflows and multi-payer statutory regulations, a prototyping paradigm will ensure system requirements are accurately captured and validated before deep technical integration. Working software increments and user



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interfaces will be continually refined through frontline stakeholder feedback and sequentially integrated with legacy hospital database feeds.

System Architecture and AI Integration (Data Privacy Compliance) To strictly comply with the Data Privacy Act of 2012 (RA 10173) and National Privacy Commission regulations, the MedCore architecture enforces an absolute architectural separation between deterministic financial calculation engines and Generative Artificial Intelligence (AI) processing. All statutory and commercial billing mathematics—including the 12% VAT exemption, 20% Senior Citizen and PWD discounts under RA 9994/RA 10754, PhilHealth All Case Rates (ACR), and private HMO benefit limits, tariff multipliers, and room-and-board caps—are executed entirely by a local, deterministic rule engine with centavo precision. Generative AI is never permitted to calculate financial balances or mutate ledger totals. Instead, AI capabilities are deployed strictly as natural-language translation, conversational guidance, clinical claims forensic auditing, and administrative automation agents.

Proposed Artificial Intelligence (AI) Architecture and System Modules

The proposed MedCore platform will implement an intelligent multi-agent Artificial Intelligence suite powered by Google AI Studio's Gemini 2.5 Flash model, selected for its rapid processing velocity, 1-million-token context window, and zero operational licensing expenditure. All AI queries will be mediated exclusively through a dedicated backend proxy (apps/server) equipped with automated local Protected Health Information (PHI) de-identification scrubbers. The proposed platform will integrate four specialized AI operational modules directly into the clinical and administrative workflows:

1. 'Cora' — Multilingual Patient Billing & Discharge Concierge Chatbot (Mobile/PWA): Cora will serve as the patient- and caregiver-facing conversational interface, accessible without client-side installation via SMS/WhatsApp magic links on standard mobile devices. Operating fluently in English, Taglish, and Ilocano, Cora will translate complex hospital chargemaster entries into structured, plain-language summaries ('Why Do I Owe This?'). The module will provide real-time discharge milestone tracking, detailing active clearance phases across attending physician professional fee signing, inpatient pharmacy reconciliation, and HMO LOA approval. For patients with remaining balances who require financial relief, Cora will provide empathetic, structured guidance on statutory patient rights under Republic Act No. 9439 (Anti-Hospital Detention Law) and facilitate configuring legally binding 0% interest digital promissory installment agreements. Furthermore, the interface will feature an integrated one-tap billing dispute mechanism that transmits contested line items directly to hospital billing supervisors for administrative review.

2. Clinical Claims Forensic Pre-Flight Scrubber & PhilHealth RTH Predictor (Staff Dashboard): Operating within the hospital staff command center, this clinical natural language processing (NLP) agent will conduct automated forensic pre-flight audits of



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PhilHealth claim packages prior to gateway transmission. The agent will ingest unstructured clinical documentation from Claim Form 4 (CF4)—including chief complaint, physical examination findings, daily course in the ward, and medication administration records (MAR)—and evaluate them against primary and secondary ICD-10 diagnostic codes, RVS procedure codes, and the machine-readable eSOA XML payload (PhilHealth Circular No. 2023-0026). The module will compute a quantitative Return-to-Hospital (RTH) predictive risk score (0.00 to 1.00), automatically flag high-risk claims exceeding a 0.15 failure threshold, identify unsubstantiated diagnoses or documentation deficiencies, and generate specific remediation recommendations to maintain clean-claim submission rates above 98%.

3. Automated HMO Medical Necessity Appeal & Dispute Letter Drafter (Staff Dashboard): To address administrative claim denials and disallowances from private HMO adjusters—specifically regarding inpatient medical sundries (PPE, surgical packs, cannulas, and restricted pharmacotherapy) and room-tier upgrade differentials—this module will automatically extract pertinent clinical indications from patient electronic records and physician progress notes. The agent will synthesize these clinical justifications to automatically draft formal, professional medical necessity appeal packages and reconsideration letters addressed to HMO medical directors, significantly reducing administrative arbitration time and mitigating inappropriate out-of-pocket shifting onto patients.

4. Executive Command Telemetry & Bottleneck Briefing Generator (Executive RCM Suite): Operating within the executive revenue cycle management (RCM) dashboard, this module will synthesize high-volume operational telemetry across the hospital's active inpatient population. The agent will continuously monitor discharge queue velocity, stage-specific latencies, Days in Accounts Receivable (DAR), and payer-specific adjudication intervals. The module will automatically generate daily executive operational briefings for hospital chief financial officers and nursing administration, identifying departments contributing to discharge delays, highlighting outstanding physician professional fee slips, and quantifying bed-hours reclaimed through automated workflows.

Proposed 5-Pillar Closed-Loop Operational Workflow

The proposed MedCore platform will operationalize hospital financial clearance through an integrated 5-pillar closed-loop operating architecture that manages the complete inpatient financial lifecycle, compressing discharge turnaround time to under 45 minutes:

1. Stage 1: GUIDE (Pre-Admission & Intake Financial Orientation): This stage will resolve upfront diagnostic and cost ambiguity by generating an automated pre-admission cost estimate based on the provisional ICD-10 diagnosis and historical hospital chargemaster data. The module will cross-reference the requested room category against the patient's private HMO daily room allowance, alerting patients to potential room-tier



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upgrade differentials and associated clinical markups prior to admission. This orientation will function as informational guidance in strict compliance with the Anti-Hospital Deposit Law (RA 10932), ensuring zero advance deposit requirements.

2. Stage 2: VERIFY (Real-Time Coverage & Multi-Payer Pre-Authorization): This stage will verify patient benefit eligibility upon hospital admission by interfacing with PhilHealth Member Data Records (MDR) to confirm qualifying contributions and determine No Balance Billing (NBB) applicability. Concurrently, the module will dispatch structured electronic pre-authorization inquiries to private HMO endpoints, establishing approved coverage thresholds and displaying a dynamic Financial Clearance Badge (Green = Verified, Yellow = Action Required, Red = Ineligible) on billing workstation consoles.

3. Core Middleware: RECONCILE (Tripartite Automated Adjudication Engine): The core middleware engine will continuously capture unadjudicated charges from legacy hospital information system (HIS) database feeds—spanning inpatient pharmacy, laboratory, central supply, and room accommodation tariffs—alongside attending physician digital professional fee entries. The engine will deterministically execute the sequential statutory deduction cascade: 12% VAT exemption, 20% Senior Citizen and PWD discounts (RA 9994/RA 10754), PhilHealth All Case Rates (ACR) allocations, and private HMO benefit limits and consumable exclusion schedules, establishing a verified net balance in real time.

4. Stage 3: REWARD / SETTLE (Statutory Settlement, Digital Promissory & Gate Pass Issuance): Upon entry of the attending physician's May Go Home (MGH) order, the system will activate a 45-minute countdown service-level agreement (SLA) timer. The patient will receive an itemized billing statement on their mobile device via SMS magic link and can complete settlement through integrated digital payment channels (QR Ph, GCash, Maya) with prompt-settlement courtesy adjustments. For patients with remaining balances who are unable to make immediate payment, the platform will automatically generate a legally binding 0% interest digital promissory note pursuant to Republic Act No. 9439, complete with captured government ID documentation and electronic signatures. Successful settlement or promissory execution will immediately issue a cryptographically authenticated Digital Discharge Gate Pass (QR Code), allowing immediate patient egress without cashier queue congestion.

5. Stage 4: REPORT (Operational Telemetry, eClaims XML & Ledgers): This stage will deliver real-time telemetry across the hospital discharge pipeline, track active SLA countdown metrics, and compile standardized PhilHealth eClaims 3.0 packages containing machine-readable XML Statements of Account (eSOA) validated against the national schema (PhilHealth Circular No. 2023-0026). The module will provide continuous tracking of essential financial performance indicators, including Days in Accounts Receivable (DAR), reclaimed bed-hours, and multi-payer accounts receivable aging.



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6. Stage 5: IMPROVE (Continuous Denial Learning & Payer Optimization): This stage will close the operational cycle by systematically ingesting PhilHealth Return-to-Hospital (RTH) notices, commercial HMO claim denials, and patient dispute logs. Utilizing Pareto root-cause analytics, the module will identify systematic error trends (such as omitted diagnostic attachments or recurring physician fee discrepancies) and automatically deploy updated validation rules across Stages 1, 2, and 3. This closed-loop mechanism will maintain claim rejection rates below 2% and equip hospital leadership with objective empirical data for annual HMO contract renegotiations.

AI Privacy-by-Design Architecture & Data Flow Pipeline (RA 10173 Compliance)

To guarantee zero leakage of Protected Health Information (PHI) under the Philippine Data Privacy Act of 2012 (RA 10173) and National Privacy Commission regulations, the proposed AI data pipeline will enforce a strict four-stage privacy protocol: (1) Local On-Premise Execution: All primary patient records and raw chargemaster entries will remain strictly within the hospital's local network infrastructure or dedicated private cloud instance; (2) Automated PHI De-Identification: Prior to payload transmission, an algorithmic scrubber will strip all direct personal identifiers (patient names, street addresses, telephone numbers, hospital case numbers, and PhilHealth PINs), transmitting only anonymized transaction IDs, standardized diagnostic classifications (ICD-10/RVS), and numerical financial quantities; (3) Secure Backend Proxy Mediation: Client applications (hospital web dashboard and mobile PWA) will be architecturally restricted from direct connections to external AI services; all requests will be routed through a secure backend proxy (apps/server at /api/v1/ai/*) utilizing mutual TLS (mTLS) and environment-level credential isolation; and (4) Non-Storage AI Processing: Inquiries processed via Google AI Studio's Gemini 2.5 Flash will execute transiently in memory under a zero-retention enterprise agreement, returning structured JSON responses that will be re-associated with patient identities strictly on the local client side.

Implementation Process

The research will operationalize the Design Science Research (DSR) framework through an integrated 9-stage engineering and evaluation lifecycle:

1. **Consultation:** Engage with hospital leadership, medical directors, nursing supervisors, and billing personnel to establish operational requirements, domain constraints, and integration boundaries.
2. **Needs Assessment:** Conduct clinical workflow analyses and baseline time-motion studies across partner hospital facilities to map discharge bottlenecks and quantify bed turnover latency.
3. **System Design:** Architect the multi-payer middleware specifications, relational database schemas, tripartite calculation models, and user interaction workflows.
4. **Development:** Build and implement the core MedCore platform, incorporating legacy HIS database connectors, deterministic deduction engines, and AI microservices.



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5. **Testing:** Execute rigorous unit testing, automated claims validation, simulated high-volume stress testing, and vulnerability auditing across all system modules.
6. **Deployment:** Configure and integrate the MedCore middleware within the partner healthcare facility operational environment for active pilot validation.
7. **Training:** Deliver comprehensive capability-building workshops and hands-on operational training for hospital billing clerks, nursing coordinators, and IT staff.
8. **Evaluation:** Conduct multi-stakeholder empirical evaluations encompassing ISO/IEC 25010 software quality benchmarks, TAM user acceptance metrics, and HGDG gender-responsiveness.
9. **Enhancement:** Implement iterative architectural optimizations based on real-world stakeholder feedback, payer policy revisions, and audit telemetry.

Criteria in the Selection and Screening of Participants

Participants will be selected through purposive sampling. To satisfy the study's dual-evaluation framework, participants will be categorized into two distinct user groups:

Group A: IT Experts and Hospital Personnel (For ISO/IEC 25010 and TAM Evaluation)

- Be current personnel or administrators of the partner regional/local healthcare facilities.
- Be directly involved in hospital IT administration, billing, PhilHealth claims submission, private HMO coordination, or clinical records management workflows.
- Be identified as potential users or administrators of the MedCore staff dashboard.
- Be willing to participate in system orientation, hands-on training, and the technical evaluation phases.

Group B: End-User Patients (For TAM Evaluation)

- Be a purposive subset of recently discharged inpatients (or their authorized family caregivers/proxies) who have experienced the hospital's discharge and billing process under PhilHealth, private HMO, or self-pay coverage.
- Have access to a standard mobile device to properly test, navigate, and evaluate the clarity of the patient-facing SMS magic links and digital checkout interfaces.

General Criteria for All Participants:

- Must provide documented informed consent to participate in the project activities.
- Personnel or patients who are not involved in the covered billing/discharge processes, or who are unavailable or unwilling to participate, will be strictly excluded from the system evaluation activities.



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Capability Building / Training

The capability-building component will ensure that designated personnel can effectively use and manage the developed systems. Training will use a hands-on approach based on actual hospital workflows and will cover:

Training Activity	Coverage	Expected Output
System Orientation	Purpose, features, functions, and user roles of the systems	Participants understand the systems and their functions
Hands-on Training	Encoding, recording, searching, updating, tracking, and report generation	Participants demonstrate basic system operation
System Management Training	Data management, basic troubleshooting, backup, and responsible information handling	Designated personnel capable of basic system management

VII. EXPECTED OUTPUTS:

Upon completion of the 12-month project duration, the research will produce:

1. **1.** A fully functional MedCore Multi-Payer Middleware Prototype deployed in a partner hospital setting, featuring: (a) an automated tripartite deduction engine (12% VAT, 20% SC/PWD, PhilHealth ACR, and private HMO eLOA rules); (b) the 5-pillar operating loop compressing post-clinical discharge clearance to <45 minutes; (c) the 'Cora' multilingual AI billing concierge chatbot (English, Taglish, Ilocano) delivering dynamic line-item explanations and RA 9439 promissory counseling via SMS magic links; (d) a clinical claims forensic scrubber predicting PhilHealth RTH denial risks against CF4 notes; (e) an automated HMO medical necessity appeal drafter; and (f) an eClaims 3.0 XML pre-flight validator.
2. **2.** A Comprehensive Impact Evaluation Report detailing ISO/IEC 25010 software quality benchmarks, TAM user acceptability metrics, Harmonized Gender and Development Guidelines (HGDG) Gender-Responsiveness certification, and quantitative statistical comparisons of discharge turnaround times (TAT) and hospital bed turnover rates against established pre-implementation baselines.



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3. **3.** Comprehensive User Manuals, Technical API Documentation, and AI Data Privacy Governance Guidelines (ensuring zero-PHI leakage) turned over to partner hospital administration and IT departments for long-term operational sustainability.
4. **4.** An Academic Capstone Research Manuscript ready for presentation and publication in high-impact national and international health informatics and computing research conferences.

VIII. SCHEDULE OF ACTIVITIES:

The implementation schedule spans a 12-month period, utilizing the developmental workplan structure to track distinct project milestones.

Major Activities Workplan

Major Activities	Q1 (Nov - Jan)	Q2 (Feb - Apr)	Q3 (May - Jul)	Q4 (Aug - Oct)
Project coordination and hospital consultation	/			
Needs assessment and process mapping	/			
Needs assessment, process mapping, and baseline data gathering	/			
System development		/	/	
System testing and debugging		/	/	
User acceptance testing and refinement		/	/	
System deployment		/	/	



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Major Activities	Q1 (Nov - Jan)	Q2 (Feb - Apr)	Q3 (May - Jul)	Q4 (Aug - Oct)
System orientation and training			/	/
System evaluation			/	/
Documentation and final reporting			/	/

Implementation Targets & Indicators

Program / Project Phase	Expected Output / Indicator	Schedule
Consultation & Design	1 partnership/MOA established System requirements specification document	Q1
Development & Testing	UAT completion report Bug-fix documentation	Q2 – Q3
Deployment & Training	Deployed systems	Q3
Evaluation & Reporting	Post-implementation monitoring report User feedback report	Q4

IX. BUDGETARY REQUIREMENTS:

(This includes a detailed/itemized breakdown of the total project cost and the source/s of funds per quarter, the first quarter should be broken down into month. (Please see attached form)



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Submitted:

FERNANDO M. JAMINOLA JR.

_____
(Printed Name and Signature of Proponent)

Endorsed by:

_____
(Signature above printed Name)